

Abderrahmen Kouki

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Professional Summary

Software engineering student specializing in Data Science and Artificial Intelligence. Hands-on experience in building ML systems, AI agents, automation pipelines, and distributed data processing (Spark/Hadoop). Skilled in RAG systems, LLM fine-tuning (LoRA, QLoRA, OOTDiffusion), MLOps (DVC, MLflow, Kubernetes) and full-stack deployment; seeking an internship in Data Science / MLOps.

Education

- **Ecole Polytechnique de Sousse – GFI** (2023–2026)
Software Engineering – Data Science & Artificial Intelligence
 - **Université Paris Dauphine – PSL** (2021–2023)
Applied Mathematics for Computer Science & Decision
 - **IPEIT Tunis** (2018–2021)
Maths–Physics Preparatory Class
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Professional Experience

DS Informatique, Paris (Remote)

Python Developer Intern

July–August 2025

- Automated factory timeclock integration, reducing payroll workload by 85%.
- Developed payroll calculation, attendance tracking, and reporting workflows.
- Built a RAG system for intelligent business report generation.

Data Soft, Sousse, Tunisia

Software Development Intern

June–July 2024

- Built an ERP extension for automated financial data transfer aligned with Tunisian tax laws.
- Implemented automated validation and reporting, reducing compliance errors by 70%.

Bureau Expert-Comptable Moncef Zayani, Tunisia

Accounting & IT Intern

July 2023

- Generated compliance reports and improved database connectivity.
 - Enhanced data visualization and anomaly detection pipelines.
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Technical Skills

- **Programming:** Python, TypeScript, Java, C#, R, React, Flask, WLanguage (WinDev)
 - **AI & ML:** Machine Learning, Deep Learning, NLP, XGBoost, RAG Systems, LLM Fine-tuning (LoRA, QLoRA, OOTDiffusion), AI Agents (LangGraph, CrewAI, Ollama)
 - **MLOps & Automation:** Docker, Podman, Kubernetes, MLflow, DVC, GitLab CI/CD, Amazon S3, MinIO, CRISP-ML(Q), FastAPI, n8n, Grafana, Prometheus
 - **Data Engineering:** Spark, Hadoop, SQL, PostgreSQL, Supabase, ETL Pipelines, Data Warehousing, Snowflake
 - **Web Development:** MERN/MEAN, GraphQL, PHP, .NET, REST APIs, Microservices
 - **Tools:** Git, GitHub, GitLab, VS Code, Jupyter
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Key Projects

Multi-Agent Debating Simulation Game (LangGraph, CrewAI, Ollama)

- Built an interactive multi-agent debating system with AI-vs-AI and Human-vs-AI modes.
- Added trending topic generation, expert personas, scoring system, and leaderboard.
- Integrated voice input and TTS for natural, game-like debate interactions.

Diabetes Risk Prediction Tool (Spark, Hadoop, XGBoost, Streamlit)

- Processed the CDC BRFSS dataset using Spark and Hadoop for distributed cleaning and feature engineering.
- Trained a calibrated XGBoost model achieving **85% accuracy**.
- Deployed a Streamlit app delivering personalized diabetes risk insights based on medical risk factors.

Gym Management & Booking System (React, Supabase, API Deployment)

- Built a full-stack gym platform for memberships, booking, attendance tracking, and payments.
- Developed Supabase schema with RLS, authentication, real-time updates, and admin dashboards.
- Deployed REST APIs and automation used daily by a local gym.

MLOps Automation Pipeline (MLflow, DVC, GitLab CI/CD, Docker, Kubernetes)

- Automated ML versioning, experiment tracking, and reproducible pipelines with DVC and S3/MinIO.
- Containerized models with Podman/Docker and deployed using GitLab CI/CD.
- Orchestrated workloads on Kubernetes for scalable inference services.

Business Analytics Dashboard (React, Node.js, Power BI)

- Built a MERN dashboard with secure authentication and backend automation.
- Integrated Power BI to deliver real-time business metrics analytics.

RAG Knowledge Bot (LangChain, FAISS, n8n)

- Developed a RAG chatbot with FAISS retrieval and automated document ingestion.
- Used n8n for scheduled syncing, preprocessing, and workflow automation.

Virtual Try-On System (OOTDiffusion)

- Implemented a virtual try-on pipeline generating realistic clothing previews.
- Integrated the feature into a mobile app for real-time visualization.

Leadership & Community Involvement

- **IEEE EPS Student Branch:** 2nd Place – IEEE Xtreme Hub Competition (2023)
- **Twice Night Challenge:** Winner – Water Footprint Reduction Project
- **Founding Member – IT Lab (Dauphine):** Organized technical workshops
- **Arim Association:** Led volunteering initiative for people with specific needs (2024)
- **ACM Member:** Participated in programming competitions

Certifications

- Microsoft Azure AI-900 – AI Fundamentals
- NVIDIA Deep Learning, NLP Transformers & Diffusion Models
- PMI Management Ready Certification
- CCNA 1 & 2 – Networking Fundamentals
- BLS – Basic Life Support (European Resuscitation Council)

Languages

Arabic: Native — **English:** Professional (C1) — **French:** Professional (C1) — **German:** Basic (A1)